

Casey Mathieson

Electro-Mechanical Systems · Automation · Precision Manufacturing

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Professional Summary

Electro-mechanical systems engineer with 11+ years of experience designing, building, commissioning, and improving automation equipment, manufacturing processes, and precision-metrology systems. Brings end-to-end ownership from root-cause analysis and concept development through mechanical/electrical design, fabrication, test, deployment, and durable process documentation. Combines hands-on manufacturing knowledge with strong mechanical design, controls, calibration, and cross-functional delivery skills.

Technical Expertise

Systems engineering & metrology: Precision metrology, laser calibration, sensors/instrumentation, motion and robotic systems, pneumatics, thermal control, test-method development, and root-cause analysis. **Mechanical design & manufacturing:** SolidWorks (CSWP), PDM, GD&T, complex assemblies, FEA/simulation, fixtures/nests, DFM/DFA, CNC/CAM/G-code, and fabrication. **Electrical, controls & functional safety:** Ground-up AC/DC panel design; schematic review/redlines; motor control/VFDs; PLCs; and machine-safety design, programming, and validation. **Commissioning, quality & process:** System commissioning, FAT/SAT, calibration, verification and validation, PFMEA, SOPs/work instructions, risk/dependency mapping, and process improvement. **AI-enabled engineering tooling & data workflows:** Agent-assisted tool development and validation; technical research, knowledge/file discovery, workflow automation, Python/MATLAB/TypeScript tools, and structured data. **Delivery & technical coordination:** Requirements definition, estimates/proposals, cross-functional coordination, customer and field support, technician-ready documentation, and knowledge transfer.

Professional Experience

ZeroTouch Metrology / DWFritz Automation — Wilsonville, OR

2020–Present

Systems Engineer, ZeroTouch Segment · 2024–Present

- Drive process improvements across precision-machine buildup, commissioning, deployment, and support.
- Conduct R&D, calibration, and test-method development for precision-metrology hardware.
- Commission systems through on-site installation and testing, resolving mechanical, electrical, and software issues.
- Develop deterministic and agent-assisted engineering workflow tools for research, SOP development, troubleshooting, and repeatable support tasks.

Mechanical Engineer · 2020–2024

- Modernized legacy machine designs and developed mechanical solutions for conveyance, packaging, part handling, and robotic end-of-arm tooling.
- Designed precision fixtures, nests, and adjustment stages.
- Improved calibration methods, DFM, CAD/BOM quality, and technician-ready documentation.
- Developed sensor- and dewpoint-based thermal-control logic to prevent condensation damage and enable safe operating responses.

Evergreen Advancements / Cascadia Digital — Oregon

2023–Present

Founder / Operator

- Build and operate a digital-product business.
- Apply automation, process design, and AI-assisted research tooling to improve repeatability in product-development and storefront workflows.

Sig Sauer Electro-Optics — Wilsonville, OR

Oct 2019–Feb 2020

Manufacturing Engineer

- Designed and integrated automated equipment and tooling for optical-electronics assembly.
- Improved reliability, throughput, and safety through equipment troubleshooting, six-axis robot programming, and PLC changes.

Production Robotics Inc. — San Leandro, CA

May 2015–Jul 2019

Senior Automation Engineer · Nov 2016–Jul 2019**Mechanical Design Engineer · May 2015–Nov 2016**

- Designed and delivered custom manufacturing and assembly automation programs ranging from \$30K to \$1.5M.
- Performed design reviews, PFMEAs, FAT/SAT, and high-risk prototype/process testing across simultaneous projects.
- Designed mechanical assemblies and ground-up AC/DC electrical panels, including motor control, VFDs, and NFPA/NEC/NEMA-aligned schematics.
- Developed estimates, proposals, schedules, and technical delivery plans for robotic assembly and material-handling systems.

West Coast Surgical LLC — Half Moon Bay, CA

Oct 2014–May 2015

CNC Machinist

- Produced high-mix, low-volume medical-device work.
- Optimized milling setups, G-code, and workholding fixtures to meet tolerance requirements.

Shaffer's Offroad — Alameda, CA

Jun 2014–Oct 2014

CAD Designer / Fabricator

- Designed and manufactured custom vehicle components using SolidWorks, CNC plasma, press-brake, milling, and MIG-welding processes.

NASA Ames Research Center — Moffett Field, CA

Jun 2013–Oct 2014

Engineering Intern, Volunteer

- Contributed to R&D for a cable-actuated tensegrity rolling robot and led a senior-project team that prototyped a modular motorized cable-actuation system.

Education & Credentials**B.S., Mechanical Engineering, Mechatronics Focus** — San Francisco State University, 2015**A.A., Economics; A.A., Marketing, Graphic Design Focus** — Santa Barbara City College, 2011**Certified SOLIDWORKS Professional — Mechanical Design** — Certificate C-QNHPJK9BLV, 2019